

Vision Language Model for Interpretable and Fine-grained Detection of Safety Compliance in Diverse workplaces

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 - 6 Vision language models
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Abstract

- PPE non-compliance causes serious workplace accidents and losses.
- Traditional detection models (**YOLO, Faster R-CNN, SSD**) struggle with fine-grained PPE attribute verification.
- VLMs link vision and text but struggle in complex scenes.
- **Clip2Safety** is proposed for interpretable PPE compliance detection.
- It has **4 modules**: scene recognition, visual prompt, detection, and verification.
- Scene recognition identifies workplace and needed PPE.
- Visual prompt generates task-specific prompts.
- Detection checks if required PPE is worn.
- Verification ensures PPE meets detailed attributes.
- Outperforms VLM baselines (**SOTA**) with **21×** faster inference.

Introduction

- Workplace safety remains a major concern across industries.
- PPE non-compliance causes frequent and severe **accidents**.
- **Manual safety checks** are slow, costly, and error-prone.
- **Sensor-based detection** is expensive and limited in scope.
- Vision models like **YOLO** and **CNNs** improve PPE detection.
- Existing methods fail to verify **fine-grained PPE attributes**.
- **Scene diversity** and **data imbalance** lower detection accuracy.
- **Clip2Safety** integrates VLMs and LLMs for smart compliance checks.
- It detects people, verifies PPE, and validates detailed attributes.
- Achieves higher accuracy and faster inference than prior models.

(a) Specifications for safety attire across various scenes are often inadequately defined

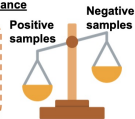


(b) Detailed criteria for identical safety items in distinct scenes frequently lack clarity



(c) Data scarcity and imbalance

CPPE-5: Medical PPE Dataset:
Over 1k images
PPEs Dataset:
Over 10k images
Safety Detection Dataset:
Over 2k images
Pictor-v3 Dataset:
Over 1k images



(d) Poor image-text embedding alignment with original images

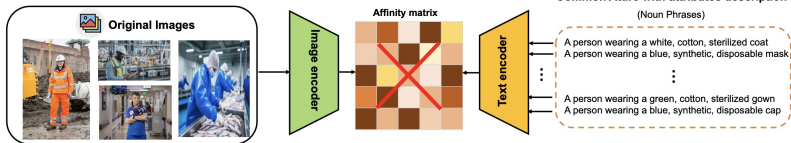


Figure 1: (a) Inadequately defined specifications for safety attire across various scenes. (b) Detailed criteria for fine-grained attributes of different safety attires vary across various scenes (c) Individuals with no safety items only take up a tiny portion of the total samples. (d) Poor image-text embedding when directly using VLMs on original images.

Figure 1. Explanation

- Figure 1 illustrates four major challenges in current PPE detection systems.
- (a) Safety attire requirements differ across scenes (hospital, factory, site).
- (b) Identical items have scene-specific attribute differences (e.g., gloves).
- (c) Datasets suffer from imbalance—few negative samples exist.
- (d) Direct VLM use causes poor image-text embedding alignment.
- These issues **reduce accuracy and adaptability** of existing models.
- Clip2Safety addresses them through **scene recognition** and better **visual-text matching**.

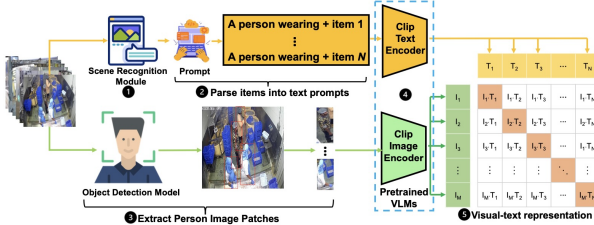
2.1 Personal Protective Equipment (PPE) Detection

- Early PPE detection used **sensor-based systems** like **RFID** and **GPS**.
- These methods were costly and required **complex hardware**.
- Vision-based systems using cameras enabled **automated detection**.
- CNN models such as **Faster R-CNN** and **YOLO** improved speed and accuracy.
- **YOLO** variants enhanced multi-scale and real-time PPE recognition.
- Most models focus on **specific scenes** and **lack general adaptability**.
- A need exists for interpretable, context-aware, and multi-scene detection.

2.2 Vision-Language Models (VLMs)

- Pre-trained VLMs like **CLIP**, **OpenCLIP**, and **Flamingo** link vision and text.
- They perform **image-text tasks** such as detection, captioning, and segmentation.
- **CLIP** models match images with text but miss fine-grained attributes.
- **VQA** models combine visual reasoning with question–answer logic.
- Datasets like **VQAv2** and **GQA** improved reasoning ability.
- Existing methods lack speed and precision for real-time PPE detection.
- **Clip2Safety** integrates VLMs and LLMs for faster, fine-grained safety compliance.

Step 1: Check if person is wearing the required safety items



Step 2: Check whether these items have the required specified features

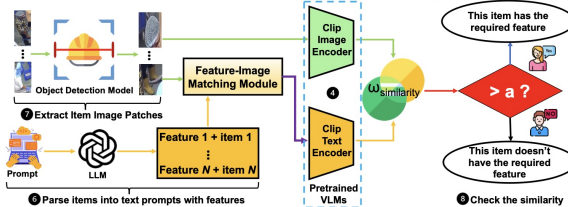


Figure 2: Model Architecture: Step 1: Detect if the person is wearing the required safety items using scene recognition and object detection, paired with VLMs for verification. Step 2: Verify that the detected safety items meet specific attribute requirements by comparing image patches with generated text prompts using a feature-image matching module.

3.1 Scene Recognition Module

- Recognizes workplace scenes to determine the required PPE set.
- Uses **BLIP2-OPT-2.7B** model for high-quality image caption generation.
- **BLIP2** combines visual and linguistic understanding for scene context.
- Employs **LoRA** fine-tuning to improve accuracy with minimal computation.
- LoRA adapts **transformer weights** using **low-rank matrices** for efficiency.
- The **loss function** minimizes negative log-likelihood of correct captions.
- Output scene captions guide the next stage: Visual Prompt generation.

3.2 Visual Prompt Module

- Uses an **LLM** to generate scene-specific PPE requirements.
- Scene captions guide the LLM to list required safety items.
- A **second prompt** generates visual features for each item.
- Example (seafood factory): hairnet, mask, gloves, apron, boots.
- Each item gets **three attributes** — color, material, and functionality.
- Example features: high-visibility, non-slip, waterproof.
- Bridges the reasoning gap between scene context and item details.
- Outputs precise visual prompts for accurate PPE detection.

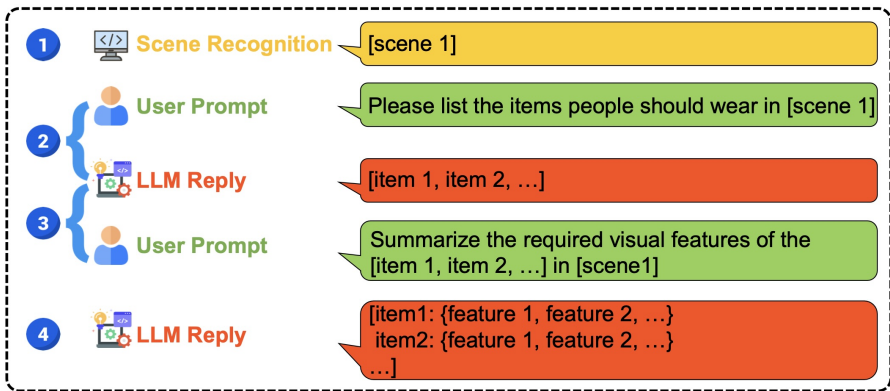


Figure 3: Clip2Safety Visual Prompt Module: Beginning with scene recognition to identify the environment, user prompts are then issued to a large language model to retrieve the necessary safety items and their specific visual features relevant to the recognized scene.

3.3 Safety Items Detection Module

- Detects whether individuals in the scene are wearing the required PPE items.
- Uses an open-vocabulary object detection model — **YOLO-World**.
- YOLO-World is pre-trained on large datasets for strong generalization.
- Generates **bounding boxes** (N_{pbbbox}) for each person in the image.
- Extracts person image patches and corresponding PPE item prompts.
- Passes patches and prompts into pre-trained VLMs like **OpenAI CLIP**.
- Produces **image embeddings** (E_{person}) and **text embeddings** (E_{item}).
- Computes an **affinity matrix** ($A_{predict} = E_{person}^T \times E_{item}$) for matching.
- Applies a **similarity threshold** (δ) to decide PPE presence.
- Optionally uses **GPT-4o** for decision refinement, improving accuracy.

3.4 Fine-Grained Verification Module

- Ensures detected PPE items meet detailed attribute requirements.
- Uses outputs from the Safety Items Detection module.
- Extracts item patches and related attribute prompts.
- Attributes include color, material, and functionality.
- Uses pre-trained VLMs to match image and text embeddings.
- Computes **cosine similarity** to verify attribute alignment.
- Applies **threshold** (τ) or **GPT-4o** for final compliance decision.
- Confirms that PPE items satisfy visual and functional standards.

3.5 Evaluation

- Evaluation measures both PPE detection and attribute verification accuracy.
- Process includes **three stages**: detection, verification, and overall evaluation.
- **Stage 1**: Detects if required PPE items are worn correctly.
- **Stage 2**: Verifies fine-grained attributes for compliance.
- Attributes grouped as DO (Direct), SO (Situational), and IO (Inferential).
- Accuracy = correctly identified items or attributes / total required.
- Combines all stages to assess complete safety compliance.

4.1 Dataset

- A combined dataset was created from four public PPE datasets.
- Sources: **Safety Detection**, **PPEs**, **CPPE-5**, and **Pictor-v3**.
- Covers **6 scenes**, **9 object categories**, and **49 attributes**.
- Scenes include hospital, construction, chemical, seafood, baking, and mechanical.
- Attributes grouped by type: **Color**, **Material**, and **Functionality**.
- Attribute levels: **DO** (direct), **SO** (situational), and **IO** (inferential).
- Enables fine-grained and multi-scene PPE compliance evaluation.



(a) Construction Site



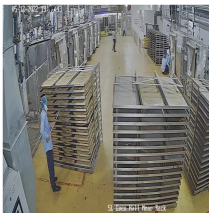
(b) Chemical Factory



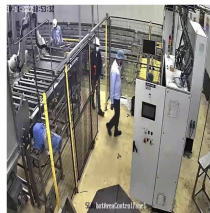
(c) Seafood Factory



(d) Hospital



(e) Baking Factory



(f) Mechanical Factory

Figure 4: Example images for 6 scenes. (a) Construction site images from Pictor-v3. (b) Chemical Factory images from Safety Detection dataset. (c) Seafood Factory images from PPEs Dataset. (d) Hospital images from CPPE-5 Dataset. (e) Baking Factory images from Safety Detection Dataset. (f) Mechanical Factory images from Safety Detection Dataset.

Table 1: **Directly Observable (DO)**: Attributes that are straightforward and can be directly observed from the visual features. **Situationally Observable (SO)**: Attributes that require some situational understanding but still have a significant visual component. **Inferentially Observable (IO)**: Attributes that are inferred based on context, usage, or additional knowledge about the object.

Type	Observable levels	Possible Values
Colors	Directly Observable	black, dark blue, dark green, dark purple, yellow, light blue, light green, light purple, white, blue, brown, green, grey, purple, red,
Materials	Situationally Observable	plastic, polycarbonate, leather, rubber, latex, nitrile, fabric
Functionalities	Inferentially Observable	shock-absorbing, impact-resistant, insulated, highly-visible, reflective, anti-slip, cut-resistant, puncture-resistant, dust-proof, fragment-proof, UV-protected, splash-proof, flame-retardant, chemical-protective, acid-resistant, alkali-resistant, face-protective, eye-protective, virus-proof, bacteria-proof, liquid-resistant, hair-covering, waterproof, contamination-preventive, stain-resistant

4.2 Implementation Details

- **Scene Recognition:** Implemented using **BLIP2-OPT-2.7B** configuration.
- **Object Detection:** Based on **YOLO-World** with `yolov8l-worldv2` setup (Ultralytics).
- **Vision-Language Model:** Used **Open-CLIP** for visual-text embedding.
- Vision encoder: **ViT-L/14@224px**; Text encoder: **OpenAI's tokenizer**.
- **LLM Integration:** **ChatGPT-4o** generates PPE items and visual attributes.
- Experiments conducted on a single **NVIDIA RTX 4090 (24GB)** GPU.
- Framework integrates multiple open-source models for efficiency and scalability.

4.3 Comparison to State-of-the-Art (SOTA) Methods

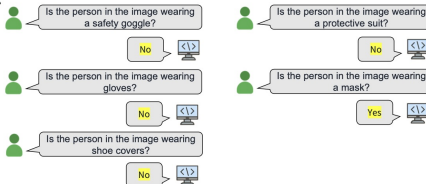
- Compared **Clip2Safety** with top VQA-based models for PPE compliance.
- Baselines: **BLIP**, **BLIP-2**, **InstructBLIP**, and **LLaVA**.
- Evaluation used **two steps** — PPE detection and attribute verification.
- Metrics: **Exact Match (EM)** and **Contains (Cont)** for answer accuracy.
- **Clip2Safety+GPT-4o** achieved the best mean accuracy of **72.3%**.
- Outperformed all baselines and was **21× faster in inference**.
- **LLaVA-1.6-7B** performed well but below Clip2Safety overall.

Visual input example of state-of-art methods:



Source: <https://sites.google.com/view/cppe5>

STEP 1



2 STEP

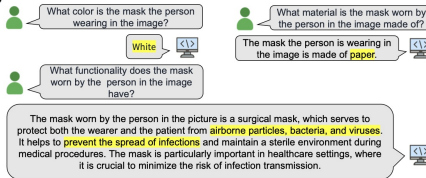


Figure 5: Example of Benchmarking LLaVA-1.6-7b for Our Safety Detection Task. Step 1 involves asking the model to identify the presence of required safety items by posing yes/no questions. Step 2 focuses on verifying specific attributes of the detected items by asking more detailed questions to ensure compliance with safety standards.

Table 2: Model details. In the prompt, “{}” is replaced by the question, QA: “Question: {} Answer:”, *: LLaVA prompt, "User: <image> Answer:".

Method	Image Res. (px)	Prompt	Visual Encoder	Text Encoder	Model Size
BILPvqa	480	{}	<i>Vit-B/16</i>	<i>BERT</i>	0.361B
BILP-2 OPT	224	QA	<i>Vit-g/l EVA-CLIP</i>	<i>unssup. OPT</i>	3.745B
BILP-2 T5	224	QA	<i>Vit-g/l EVA-CLIP</i>	<i>inst. FlanT5 XL</i>	3.942B
InstructBILP T5	224	QA	<i>Vit-g/l EVA-CLIP</i>	<i>inst. FlanT5 XL</i>	4.023B
InstructBILP V	224	QA	<i>Vit-g/l EVA-CLIP</i>	<i>Vicuna-7B</i>	7.913B
LLaVA-1.5-7B	224	*	<i>ViT-L/14 CLIP</i>	<i>LLaMA 7B</i>	6.743B
LLaVA-1.6-7B	224	*	<i>ViT-L/14 CLIP</i>	<i>Vicuna-7B</i>	7.06B
Clip2Safety	224	/	<i>ViT-L/14 CLIP</i>	<i>OpenAI's tokenizer</i>	0.428B

Table 3: Accuracy of our proposed Clip2Safety method compared to other state-of-the-art methods, evaluated step by step.

Method	Step 1	Step 2			Mean
		DO	SO	IO	
BILP vqa	76.4	61.6	53.2	3.2	48.6
BILP-2 OPT	72.0	39.6	58.0	19.2	38.2
BILP-2 T5	78.4	48.4	32.4	39.2	49.6
InstructBILP T5	76.8	52.8	39.2	29.2	49.5
InstructBILP V	81.6	46.4	60.0	5.2	48.3
LLaVA-1.5-7b	74.4	54.8	62.8	82.0	68.5
LLaVA-1.6-7b	91.2	48.4	62.0	78.8	70.1
Clip2Safety	76.8	76.9	61.4	65.8	70.2
Clip2Safety+GPT4o	77.2	79.7	66.6	65.8	72.3

4.4.1 Object Detection Model Comparison

- Ablation study compares **SSD**, **OWL-ViT**, and **YOLO-World**.
- Baseline: direct detection without object detection model.
- **SSD**: improved from 63.4% to 68.1% with GPT-4o.
- **OWL-ViT**: rose from 59.6% to 71.3% using GPT-4o.
- **YOLO-World**: achieved best accuracy — up to **72.3%**.
- Dynamic threshold (δ) tuning further enhanced performance.
- **YOLO-World + GPT-4o** gave highest accuracy and model stability.

Table 4: Inference time of our proposed Clip2Safety method compared to other state-of-the-art methods, evaluated step by step.

Method	Step 1	Step 2			Mean
		DO	SO	IO	
BILP vqa	1.3	1.3	1.3	1.6	1.4
BILP-2 OPT	7.9	7.0	8.3	12.0	8.8
BILP-2 T5	6.1	6.0	7.5	8.2	7.0
InstructBILP T5	13.9	9.9	12.7	9.9	11.6
InstructBILP V	48.6	49.7	43.2	46.6	47.0
LLaVA-1.5-7b	51.9	60.5	107.0	123.2	85.7
LLaVA-1.6-7b	140.9	144.1	202.3	237.6	181.2
Clip2Safety	0.4	2.8	2.1	2.9	2.1

Table 5: Ablation study of different object detection models. (*OD represents Object Detection Model)

Method	Step 1	Step 2			Mean
		DO	CO	IO	
(a) without OD	60.8	60.4	65.1	51.2	59.4
(b) SSD	72.8	73.0	53.1	54.5	63.4
(c) SSD+diff δ	72.8	62.8	63.1	53.5	63.1
(d) SSD+GPT4o	76.8	77.6	64.3	53.5	68.1
(e) OWL-ViT	72.8	52.9	54.3	58.2	59.6
(f) OWL-ViT+diff δ	72.8	73.6	64.7	59.8	67.7
(g) OWL-ViT+GPT4o	79.2	79.3	64.7	61.8	71.3
(h) YOLO-World	76.8	76.9	61.4	65.8	70.2
(i) YOLO-World+diff δ	76.8	78.8	63.8	62.2	70.4
(j) YOLO-World+GPT4o	77.2	79.7	66.6	65.8	72.3

4.4.2 Pretrained VLM Comparison

- Compared pretrained **CLIP** and **SigLIP** models for accuracy and consistency.
- **CLIP-ViT-L-patch14**: best performer with 70.4% mean accuracy after tuning.
- **CLIP-ViT-B** models improved from 44% to 57–63% with threshold adjustment.
- **SigLIP** variants achieved 55–57% accuracy, showing less sensitivity to tuning.
- Larger patch models benefited more from dynamic thresholding.
- Overall, **CLIP models outperformed SigLIP** in accuracy and adaptability.

Table 6: Ablation study of different VLMs. Methods include variations of CLIP and SigLIP models with different model size, patches and threshold values.

Method	Step 1	Step 2			Mean
		DO	CO	IO	
(a) CLIP-ViT-L-patch14	76.8	76.9	61.4	65.8	70.2
(b) CLIP-ViT-L-patch14+diff δ	76.8	78.8	63.8	62.2	70.4
(c) CLIP-ViT-B-patch16	73.2	26.0	29.5	49.0	44.4
(d) CLIP-ViT-B-patch16+diff δ	61.6	61.3	58.0	47.6	57.1
(e) CLIP-ViT-B-patch32	73.2	26.0	29.6	48.5	44.3
(f) CLIP-ViT-B-patch32+diff δ	52.8	74.1	64.2	62.9	63.5
(g) SigLIP-base-patch16-384	26.8	74.0	70.4	51.0	55.6
(h) SigLIP-base-patch16-384+diff δ	36.4	61.3	48.8	35.3	45.5
(i) SigLIP-so400m-patch14-84	26.8	74.0	70.4	51.0	55.6
(j) SigLIP-so400m-patch14-84+diff δ	52.8	74.2	57.8	43.6	57.1

4.4.3 Threshold Value Comparison

- Studied the impact of different threshold (δ) values on accuracy.
- Range 0.58–0.65 tested across detection and verification stages.
- **ROC curves** illustrate the trade-off between TPR and FPR.
- **Step 1** showed best performance with **AUC = 0.76**.
- **DO** step had lower AUC (0.64) due to higher visual complexity.
- **SO** step remained robust with AUC = 0.76.
- **IO** step was weakest (AUC = 0.55), hardest to infer attributes.
- Higher thresholds cut false positives but reduced sensitivity.
- Adaptive threshold tuning yielded optimal model performance.

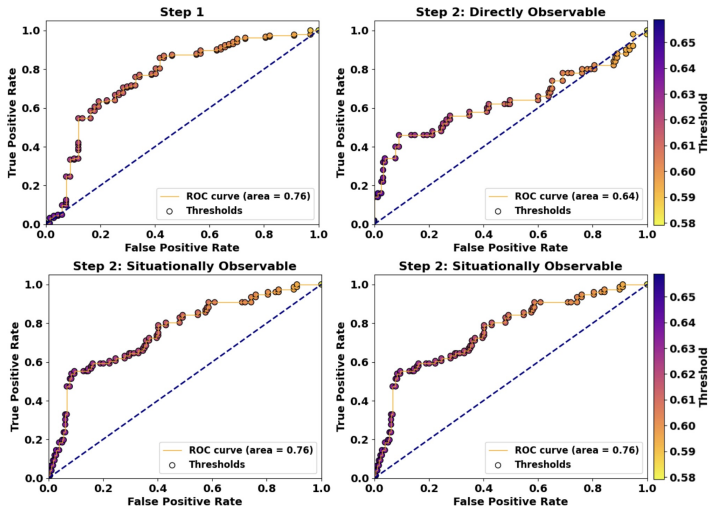


Figure 6: ROC curve analysis for different steps and threshold values. These plots show the true positive rate versus false positive rate for each step. The solid blue line represents random performance and circles indicate different threshold values, with their colors corresponding to the specific threshold level. The area under the curve values highlight the model's effectiveness in each step.

4.5 Visualization and Discussion

- Figure 7 shows visual results of the **Clip2Safety** framework.
- **Red boxes** mark people; blue boxes mark detected PPE items.
- **(a)** Successful detection and verification in both steps.
- **(b)** Correct PPE detection but attribute verification failed.
- **(c)** Weak detection in complex and cluttered environments.
- Highlights sensitivity to lighting, pose, and scene variability.
- Demonstrates strong overall performance with scope for refinement.

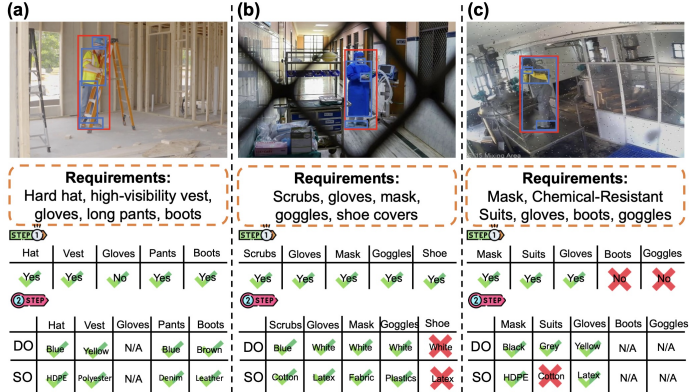


Figure 7: Visualization for detection results of the Clip2Safety. (a) Example with good performance on both step 1 and step 2. (b) Example with good performance on step 1 and bad performance on step 2. (c) Example with bad performance on step 1. (Here we only show DO and SO in step 2).

5. Conclusions

- **Clip2Safety** is an interpretable AI framework for PPE safety compliance.
- Combines **CLIP**, image captioning, and LLMs for scene-based reasoning.
- Handles diverse PPE needs through visual-text understanding.
- Uses open-vocabulary detection for accurate embedding alignment.
- Achieves higher accuracy and faster inference than existing models.
- Outperforms VQA-based methods in detection and verification tasks.
- Enables fine-grained, context-aware workplace safety analysis.